RESEARCH BRIEFING · NOVEMBER 2026

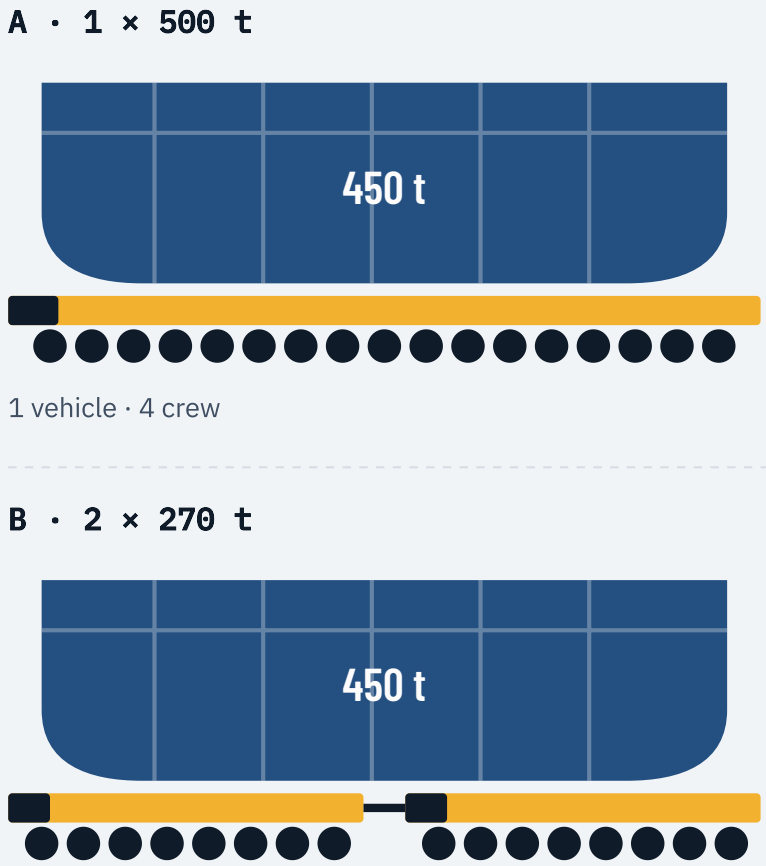
One big transporter for a heavy hull block, or several small ones coupled?

Should a shipyard buy transporters large enough to carry each block alone, or cheaper small ones that join up for heavy blocks? This study compares the cost of the two choices at the same level of service.

Service-constrained fleet sizing for single-carrier and cooperative block transport in shipyards

> A block is a piece of a ship's hull weighing tens to hundreds of tons. Today's question: is one large vehicle or several coupled small ones cheaper?

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The paper is complete and being prepared for submission



The study at a glance

1

The question

Coupled transporters can carry blocks too heavy for one. So should a yard buy heavy units that carry every block alone, light units that couple for heavy blocks, or a mix?

2

What was done

Every fleet first meets the same service: at least 95% of blocks on time over 30 days and none more than 2 h late. At its smallest qualifying count, its daily cost follows from prices and wages.

3

What was found

In **91%** of 1,620 cost comparisons under the main labor count, the cheapest fleet couples at most one block in ten; 100% where one size carries every block alone.

4

Why

Coupling ties up two or three vehicles and their crews for the whole trip, plus joining time. And people cost more than vehicles: a vehicle's daily purchase cost is below one crew's daily wages.

Coupling is a remedy for exceptional blocks, not a purchasing strategy.

In short: we made every fleet equally punctual, then compared daily costs. The cheapest fleet almost always uses one vehicle per block, because coupling ties up several vehicles and crews at once, and in a shipyard people cost more than vehicles.

Sources: workload, block masses and handling times from published studies (see References). Prices: 33 public award notices (2022–2026, three procurement platforms). Wages: National Bureau of Statistics of China. Boxes 3 and 4 are results of this study.

The order of this briefing (click to jump)

1 **The problem**
How ships are built, how blocks move, what coupling is — p. 4

2 **Approach**
How fleets are compared fairly and how costs are counted — p. 10

3 **Findings**
What the cheapest fleet looks like — p. 16

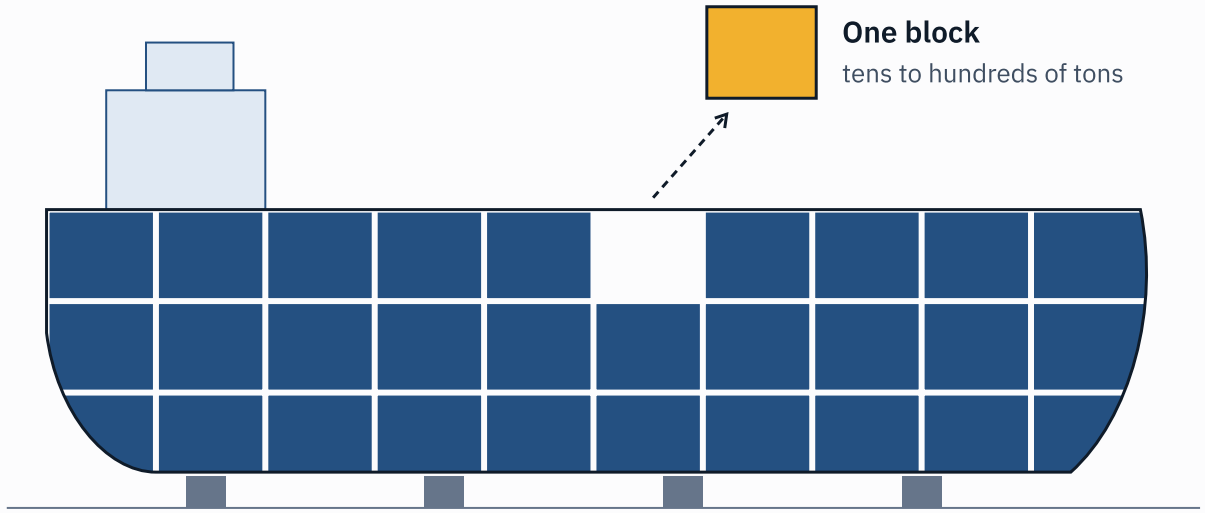
4 **Why**
Why coupling costs more — p. 20

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How the findings were checked — p. 21

6 **Advice and conclusions**
When coupling helps, advice for shipyards — p. 23

First, where the problem comes from. Then how we compare fleets fairly. Then the findings, the reasons and the checks. Last, our advice for shipyards.

Ships are built in blocks, then joined in the dock



A hull in the dock, welded together from many blocks (schematic)

- A large ship is divided into many **blocks**. They are built, outfitted and painted **in parallel** in different shops. Then they are joined in the dock.
- A typical block is about 15 m × 15 m × 5 m. It weighs 100–300 t, and some exceed 500 t.
- The price: blocks are **moved many times**. A late move holds up the next process.

This study is about those moves. Which transporters, and how many, move them at least cost?

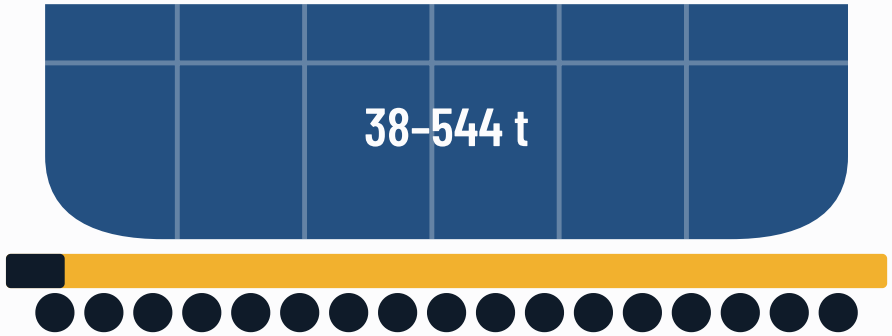
It is like building with toy bricks. Many blocks are made at once in different workshops, then joined in the dock. Work runs in parallel, but every block is moved many times. If one move is late, the next job waits.

Sources: block size and mass Park & Seo (2012), p. 390.

Each block is moved several times between shops, yards and the dock

A typical route of one block

Assembly shop → Outfitting shop → Paint shop → Block yard → Assembly yard → Dock



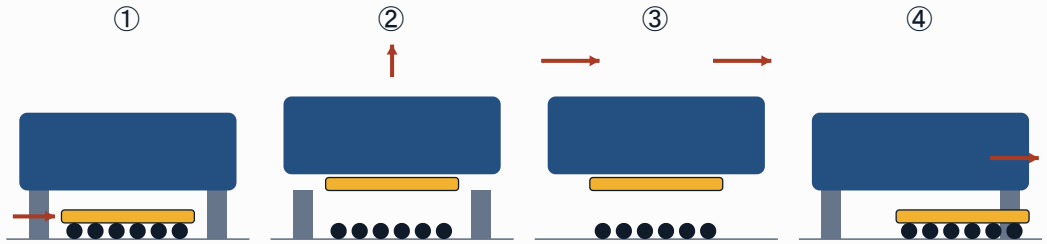
Self-propelled hydraulic transporter, rated 200–550 t

- **Block mass**: 38–544 t in published samples
- **Transporter**: self-propelled hydraulic platform, sold by rated capacity from 200 to 550 t
- **Price**: about CNY 1.79–3.92 million each before tax, fitted to public award prices
- **Crew**: one driver and three guides per vehicle, four in all
- **Workload**: on one real working day, 4 vehicles moved 97 blocks in 16 h

A block leaves the workshop, gets equipment and paint, waits in the yard, is joined with other blocks, and finally enters the dock. Every leg needs a transporter. All numbers on this page come from public sources.

Sources: masses Jiang et al. (2021) Tables 5, A1, A2. Liu et al. (2022) S1 data. Roh & Cha (2011) Table 2. Fleet ratings Jiang et al. (2021) Table 5. Roh & Cha (2011) Table 1. Liu et al. (2022) Table 4. Crew Park & Seo (2012), p. 392. Workload Roh & Cha (2011), p. 3246. Prices 33 public tender awards (2022–2026).

How a transporter moves a block: drive under, lift, drive, set down



① Drive in empty

The block rests on supports. The vehicle drives into the gap underneath

② Lift hydraulically

The platform rises and lifts the block off the supports. This is loading

③ Drive loaded, slowly

It carries the block to the destination. It drives at about half its empty speed

④ Set down, leave

The platform lowers the block onto new supports: unloading. The empty vehicle heads for the next block

Rated capacity

Sold by tier. One unit cannot carry a block above its rating

Speed

270 t model: 12 km/h empty, 6 km/h loaded

Loading / unloading

10 min each. A mass-dependent formula also exists. Loading in practice often takes 30–60 min

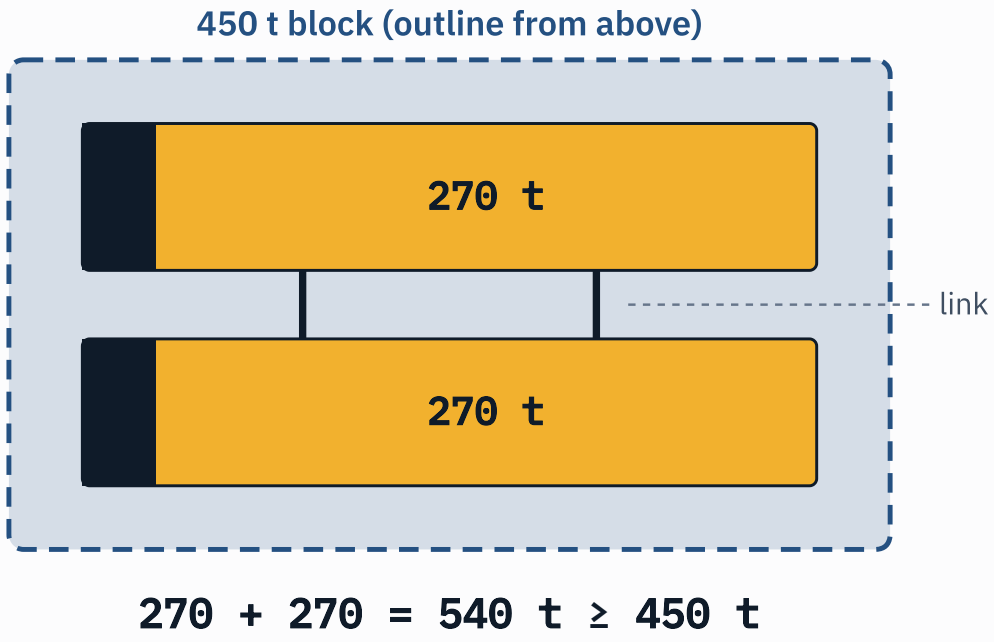
Crew

Four or five per vehicle in practice: a driver and guiders. Here each has its own crew of four: one driver, three guiders

The vehicle drives under the block, lifts it, drives slowly to the target and sets it down. Loaded, it moves about 6 km/h, like a brisk walk. Two rules: a vehicle cannot carry a block above its rating, and each vehicle keeps its own crew of four.

Sources: speeds from the Suzhou Dafang DCY270 data sheet (accessed 2026-09-24). 10 min each: Liu et al. (2022) §4.1.1. Mass-dependent formula: Kim & Joo (2012) §4. 30–60 min: Liu et al. (2022) §4.2.2. Crew: Park & Seo (2012), p. 392.

Coupling: two or three units side by side drive as one vehicle



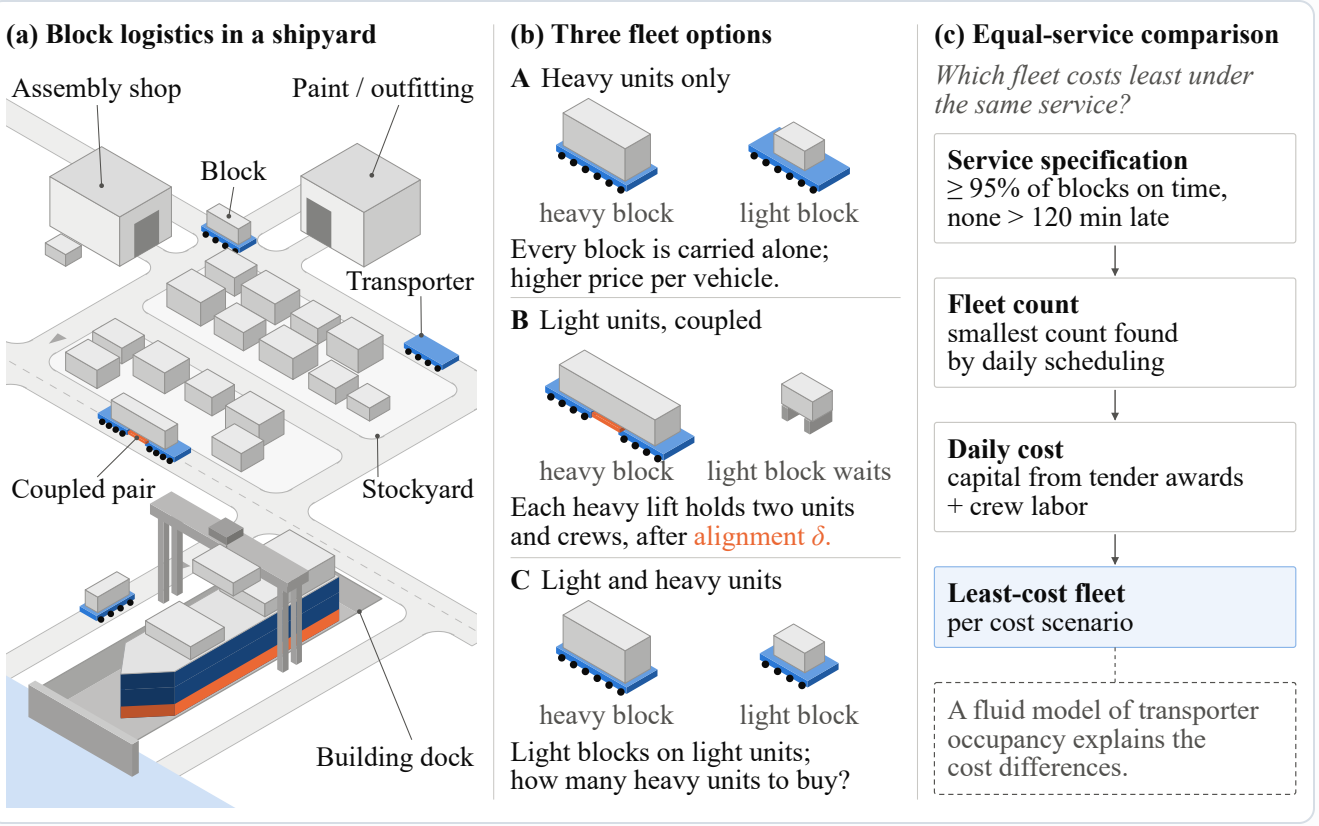
- **Capacities add**: two 270 t units lift up to 540 t, three up to 810 t. The team runs under synchronized control
- **All must be there**: the first to arrive waits for the last
- **Alignment takes time**: no measured value is published. We assume 10 min and test 0–40 min
- **Held throughout**: from loading to unloading, each unit and its crew can do nothing else
- At most three units per team (assumed). Teams of at most two were also tested

Coupling lets light units carry heavy blocks. But one block then takes two or three times the vehicles and crews. It saves capital and costs vehicle-time.

Think of a heavy table. One strong person can carry it alone. Two ordinary people can carry it too, but both must show up and both are busy for the whole trip. Coupling saves the price of a large vehicle but uses more vehicle time and labor.

Sources: synchronized control, TII Group self-propelled transporter brochure (0317-EN), p. 21. Big tasks by several flatcars, Jiang et al. (2021), constraint (24). Coupled-use orders on the Suzhou Dafang website. Alignment time and team size are our assumptions.

The problem in one figure: block logistics, three fleet options, one comparison



- ① **Block logistics**: every block is moved several times between workshops, yards and the dock
- ② **Three fleets**: A buys only heavy units and carries each block alone. B buys only light units and couples them for heavy blocks, which takes time to join. C is mainly light units plus a few heavy ones for the heavy blocks
- ③ **Service first, then cost**: every fleet first meets the same level of service, then daily costs are compared

> **Every fleet is held to the same service before costs are compared.**

The three fleets in the figure are the three ways a shipyard can buy. Every comparison that follows asks the same question: at equal punctuality, which way costs the least per day?

Sources: paper Fig. 1 (schematic). The yard is generic, not the layout of any real yard.

Three questions this study answers

(1) What to buy

At the same level of service, which size, how many units, and whether to mix light and heavy give the lowest daily cost?

(2) When coupling pays

Coupling saves the price of large vehicles but ties up more vehicles and people. Under what workload and prices can it beat buying large vehicles?

(3) How to define service

Only an average on-time rate, or also a limit on how late any single block may be? Does the answer change with the definition?

The gap: in the studies reviewed, earlier work schedules transport for a given fleet or compares how many vehicles are needed. None compares the two purchasing choices at the same level of service: large vehicles that carry blocks alone, or light vehicles that rely on coupling.

A shipyard must answer all three before buying, and the last slide answers each one. In the studies we reviewed, most plan transport for a fixed fleet. None compares the two ways of buying at equal punctuality.

Sources: the full references are on the last pages.

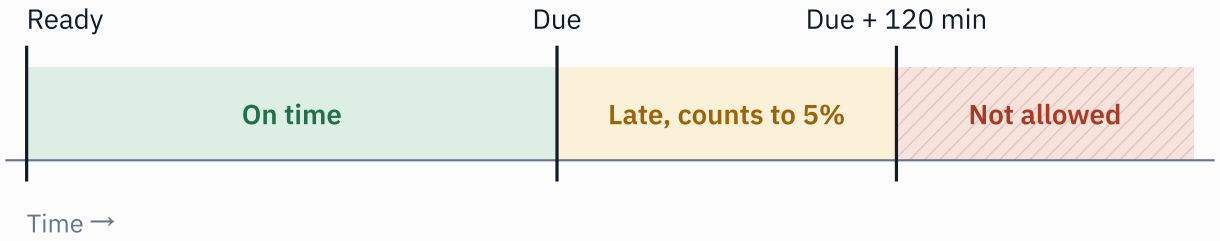
The core idea: equal service first, then cost

On time ≥ 95% · each block ≤ 120 min late

An analogy

Two bus fleets cannot be compared on price alone. Ten buses run 80% on time, and twenty minibuses run 99%. First give both the same punctuality. Then see which is cheaper.

- **Step 1**: build a timetable for each of 30 days for every fleet: which vehicle carries which block, and when
- **Step 2**: find the smallest number of vehicles that still meets the service level
- **Step 3**: at that count, compute the daily purchase and labor cost



- **95% on time**: of 2,910 blocks in 30 days, at least 2,765 must be on time
- **120-min limit**: no block may be more than 2 h late. That is the gap between two batches, so a late block never spills past the next batch

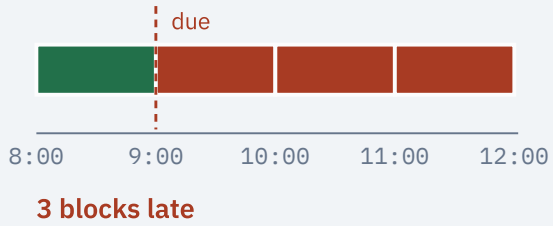
The core idea: equal punctuality first, then compare price. The rule has two parts: 95% of blocks on time, and no block more than 2 hours late. Two hours is the gap between batches, so a late block never spills into the batch after next.

Sources: the 95% target and the 120-min limit are set by this study, as the literature has no common standard. The limit was also tested from 30 min to none, and the target at 90% and 98%.

Why not divide total workload by hours per vehicle?

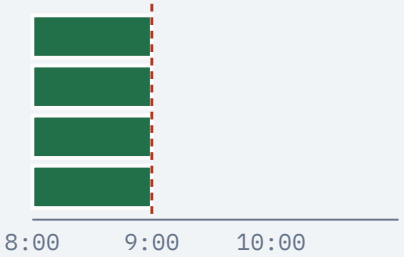
Example: four blocks ready at 8:00, all due by 9:00, each move 1 h

Totals only: 4 vehicle-h ÷ 16 h = 0.25 → one vehicle?



3 blocks late

By due time: four vehicles at once



All on time, but it takes 4 vehicles

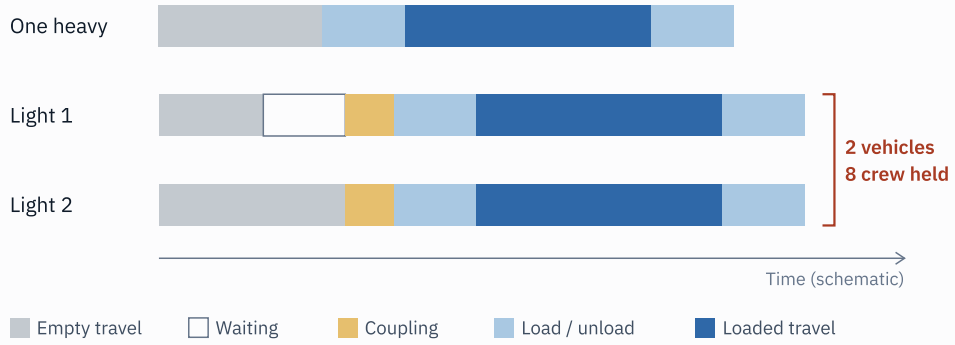
- **Work arrives in batches** with their own due times, so demand is uneven
- **Vehicles drive empty between tasks**, and the order changes that distance
- **A coupled lift needs several free vehicles at once**. One late arrival holds the team.

So a real timetable has to be built for every day to know how many vehicles a fleet needs.

By total work, four blocks are only four hours of work, so one vehicle seems enough. But all are due by 9:00, so four vehicles must work at once. Vehicle counts cannot come from simple division. They must come from daily timetables.

Source: an illustrative example.

The price of coupling: vehicles must meet and are held together throughout



One move, close up

- **Carried alone**: the vehicle loads on arrival and leaves for the next job as soon as it unloads
- Example: a 420 t block carried alone by one 425 t unit uses 43 vehicle-minutes. Two coupled 270 t units use 110, two and a half times as much

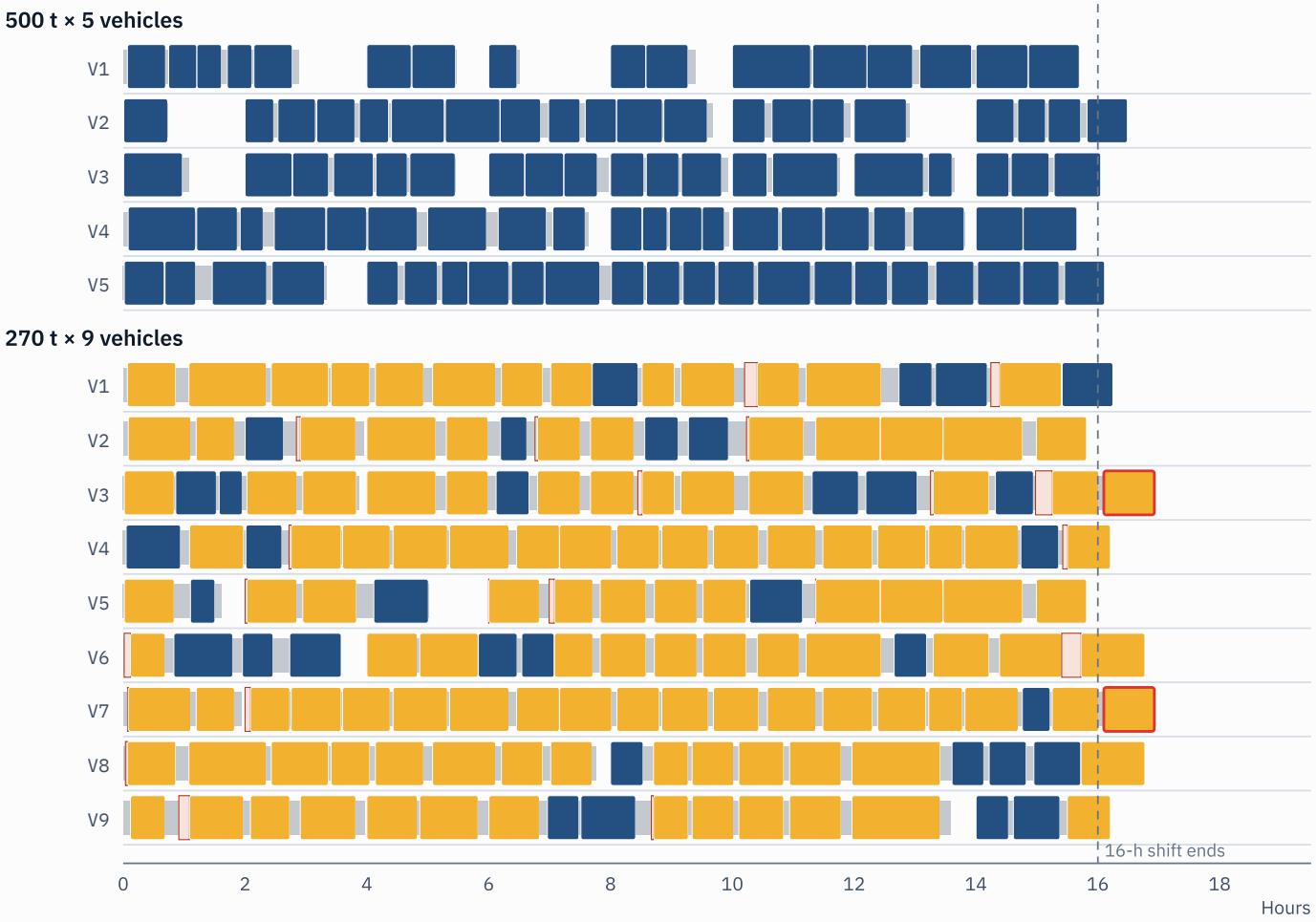
Vehicle-minutes = number of vehicles × minutes they are held.

- **Coupled**: the first vehicle waits for the second (white), then both align (yellow)
- A light block that waits for one of those vehicles is done only at minute 98, against minute 43 with single carriage

Alone, a vehicle loads as soon as it arrives. Coupled, the first vehicle waits, then both line up. For the same 420-ton block, one vehicle is busy 43 minutes, while two coupled vehicles are busy 110 minutes in total, 2.5 times as long. Other blocks wait too.

Source: illustrative times. Single carriage 43 min = 10 min empty travel + 33 min to load, carry and unload. The 10-min alignment is set by this study.

The same day's 97 blocks given to two fleets: real timetables



	500 t × 5	270 t × 9
Vehicles	5	9
On time that day	97 / 97	96 / 97
Blocks coupled	0	62
Vehicle work (vehicle-h)	70.0	145.0
· empty travel	5.7	16.7
· waiting for team	0.0	1.9
Latest block's delay (min)	0	26
Fleet price (in 270 t units)	8.15	9.00
Daily cost (labor-hours)	421	688

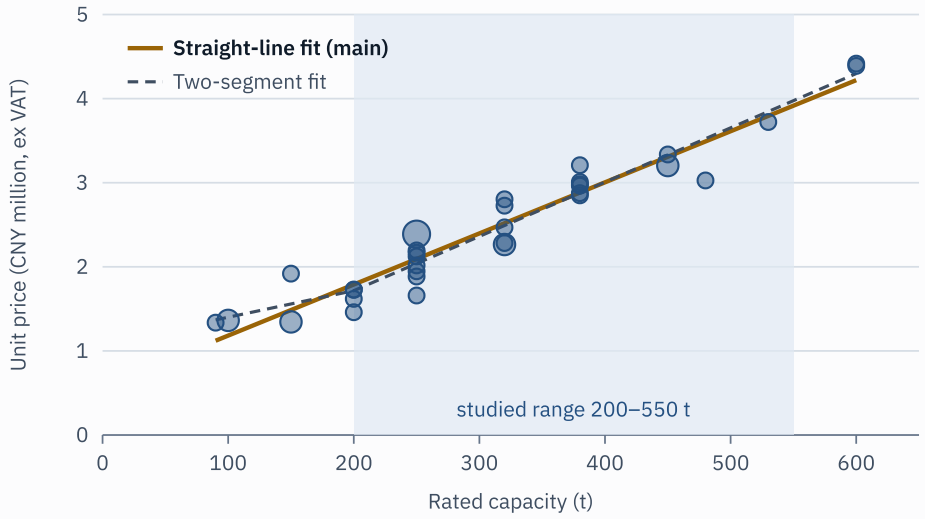
Five 500-ton vehicles are enough, with no coupling. The 270-ton fleet needs nine, couples 62 blocks and doubles its vehicle hours. With labor, it costs over 60% more per day, and even its vehicles cost more to buy.

Each row is one vehicle, and each bar is what it is doing. Hover over a bar to see the block. Both timetables are at each fleet's smallest qualifying count.

Source: timetables from this study. Day 1 with uniform masses, short handling and standard due dates. Daily cost follows the next page, with a vehicle-to-wage price ratio of 12.4.

A vehicle’s daily purchase cost is below one crew’s daily wages

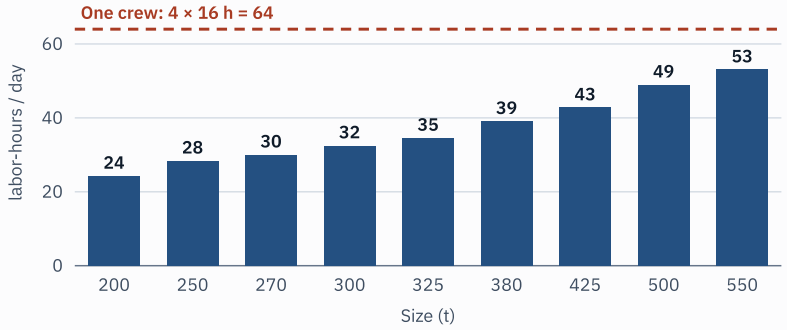
Award prices of 39 transporters (33 notices, 2022–2026, before tax). Shaded: the 200–550 t studied here.



Larger units cost more, but price grows more slowly than capacity: a 550 t unit costs 1.7–2.0 times a 270 t unit for 2.04 times the capacity. In numbers, a 270 t unit costs about CNY 2.215 million; spread over 8 years at 4% interest, that is about CNY 1,316 per working day. A worker costs about CNY 51.5 an hour with social charges. So a vehicle’s daily purchase cost is about 26 labor-hours, while a crew of four working 16 h is 64.

Daily purchase cost per vehicle (bars) against one crew’s daily wages (dashed), in labor-hours.

Vehicle-to-wage ratio 30.0



At a vehicle-to-wage ratio of 30.0, the dearest unit, 550 t, costs 53 labor-hours a day to buy, still below one crew’s 64 labor-hours a day. So one vehicle fewer saves more than moving up a size costs.

Vehicle-to-wage ratio: a 270 t unit’s daily purchase cost in labor-hours. It is about 25.6 at national wages and 11.0 at Shanghai wages, and 5.1–30 over other loan terms and interest rates. Every result was tested over this range.

We put prices and wages in one unit, the labor-hour: one worker's wage for one hour. A 270-ton vehicle costs about 26 labor-hours a day to buy. Its crew of four costs 64. So one vehicle fewer saves more than moving up a size costs.

Sources: 33 award notices from csscbidding.com (26), 365trade.com.cn (5) and e.sinochemitc.com (2). Wages: 2025 average annual wage of production workers in manufacturing enterprises above designated size, National Bureau of Statistics of China; Shanghai wages from the China Statistical Yearbook 2025; employer charges at the official social-insurance rates. Fits and conversions by this study.

Scale of the study: 36 production conditions, 30 working days each

- **6 block-mass profiles**: five rest on published samples, and one is a stress test with extra-heavy blocks (9% above 550 t)
- **3 handling times**: short, mass-dependent and long loading, all from the literature
- **2 due-date levels**: standard (from the literature) and tight
- **11 fleets**: nine single sizes (200–550 t) and two mixes of light units with one or two heavy units
- **Road network**: the digital layout of a large shipyard (not public). A second yard uses public data from a Shanghai cruise-ship yard

36

production conditions = 6 masses × 3 handling times × 2 due-date levels, 30 working days each

384

fleets with their counts determined

6,480

cost comparisons (180 cost settings per condition)

214,210

scheduling runs, all reproducible

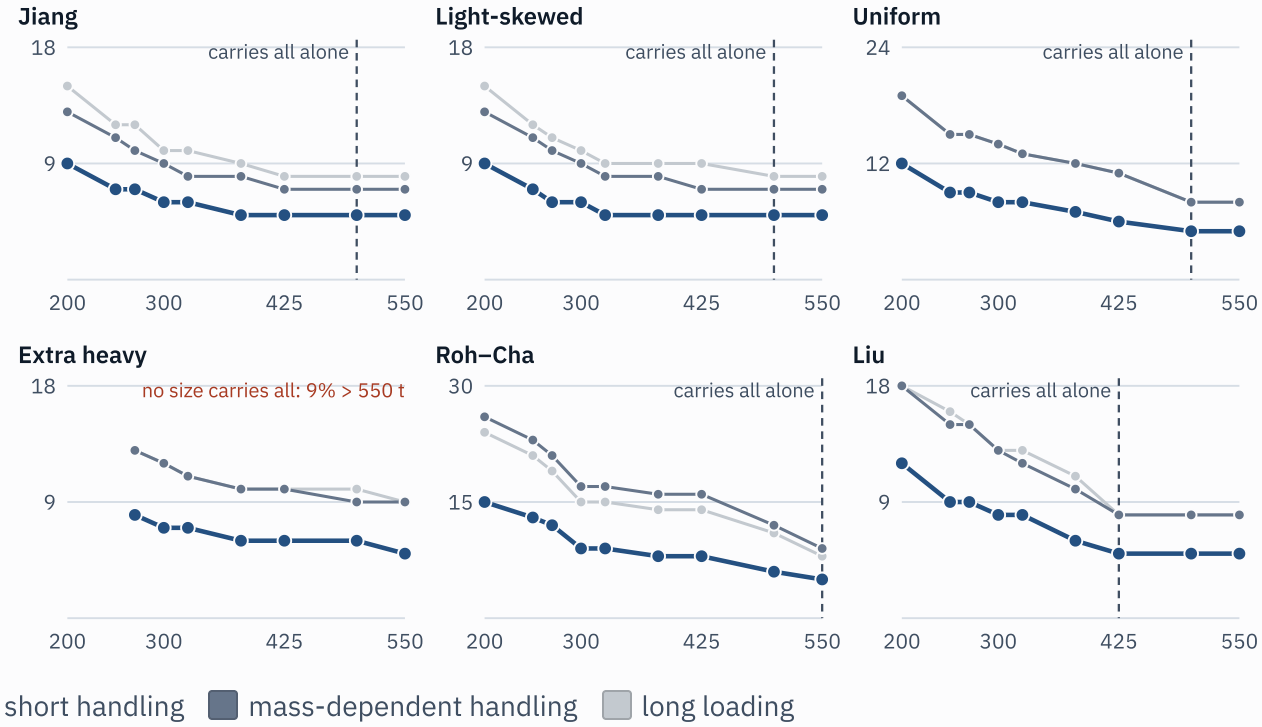
180 cost settings = 5 price fits × 9 vehicle-to-wage ratios × 4 ways of counting labor. The main labor count is full-shift crews: four people per vehicle for the whole 16 h.

We used a wide range of conditions to see whether the findings hold beyond one case. One weight case is a deliberate stress test: 9% of blocks exceed 550 tons, so no single size can carry every block alone.

Sources: block masses, handling times and due dates from Jiang et al. (2021), Roh & Cha (2011), Liu et al. (2022), Kim & Joo (2012), Park & Seo (2012) and Kweon et al. (2026). Run counts by this study.

Finding 1: larger vehicles mean fewer units, up to the size that carries every block

Smallest count that meets the service level (standard due dates). Dashed: the smallest size that carries every block alone.



- Take the Liu masses with short handling: 200 t needs 12 units, 270 t needs 9, 380 t needs 6, and 425 t or more needs 5
- Slower handling needs more units, with the same pattern
- Above the size that carries every block alone, larger vehicles save no further unit
- Below that size, heavy blocks must be coupled, and the masses decide how many: 200–380 t fleets couple 5–100% of blocks

> **The extra units of a light fleet are the price of coupling.**

Larger vehicles mean fewer vehicles, but only until one vehicle can carry every block alone. Beyond that size, a bigger vehicle saves nothing. Below it, heavy blocks must be coupled. The extra vehicles in a small fleet are the price of coupling.

Source: this study, 36 production conditions × 11 fleets.

Finding 2: the cheapest fleet carries almost every block alone

91%

of 1,620 comparisons with full-shift crews: the cheapest fleet couples at most 1 block in 10

100%

of the 1,350 where one size carries every block alone

57.5%

of all 6,480 comparisons: the cheapest fleet couples no block

24.1%

of all comparisons: won by light units plus 1–2 heavy units, coupling at most 1 in 10

Share where the cheapest fleet couples at most one in ten, by labor count (1,620 comparisons each)



* Assumes a whole coupled team needs only one crew, the most favorable case for coupling.

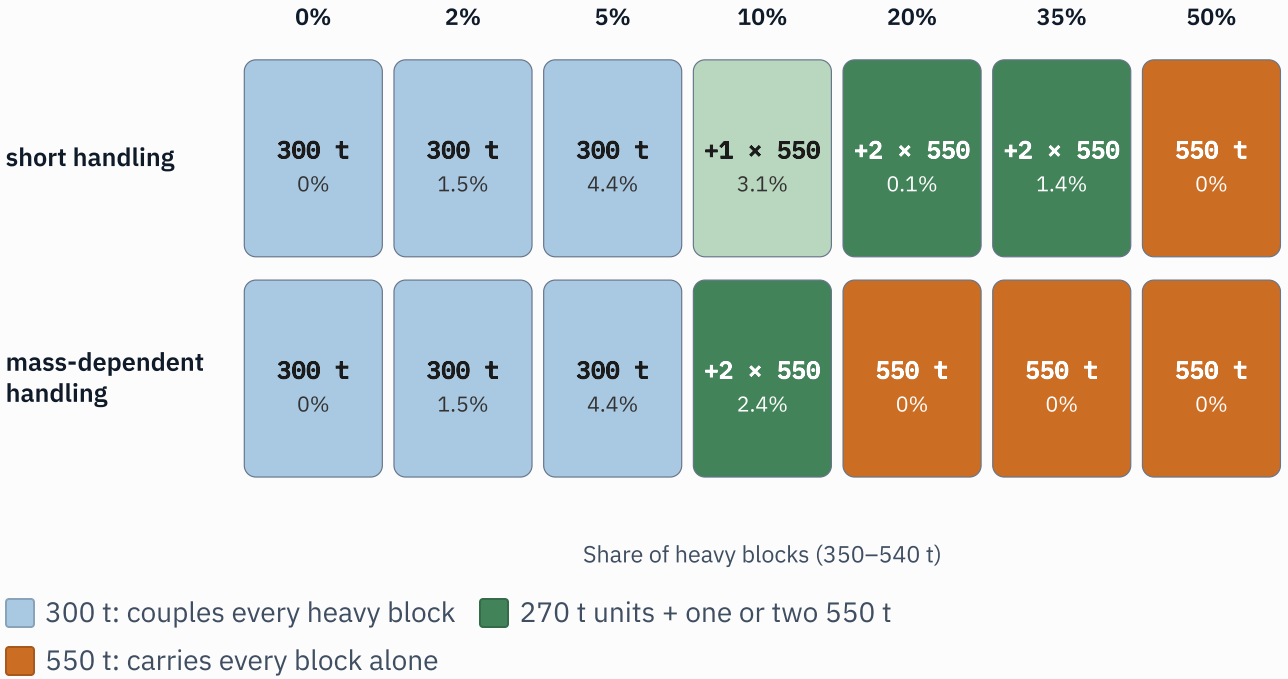
- A 15% threshold raises the share to 97.3%; the 5 price fits give 90.0–91.6%
- The median gap between first and second place is 2.4%, so the paper also reports near-cheapest fleets

All four ways of counting labor agree: the cheapest fleet carries almost every block alone. The way most favorable to coupling assumes one crew per coupled team, and even then it is 86.9%. The top options are often close, so we list several.

Source: this study. 6,480 cost comparisons = 36 production conditions × 5 price fits × 9 vehicle-to-wage ratios × 4 ways of counting labor.

Finding 3: more heavy blocks make the cheapest fleet heavier, yet it still rarely couples

Cheapest fleet at each share of heavy blocks (350–540 t); the rest weigh 100–300 t. The number in each cell is the share of blocks coupled.



- With no heavy blocks, the 300 t fleet is cheapest
- From a 5–10% heavy share, light units with one or two heavy units are cheapest
- From a 20–50% heavy share, depending on handling, an all-550 t fleet is cheapest
- At every share, the cheapest fleet couples 3.3% of blocks or fewer on average

2,502 / 2,520

comparisons where the cheapest fleet couples at most one block in ten. The exception is light units with two heavy units (19.6% coupled), at a 35% heavy share with mass-dependent handling and one crew per coupled team.

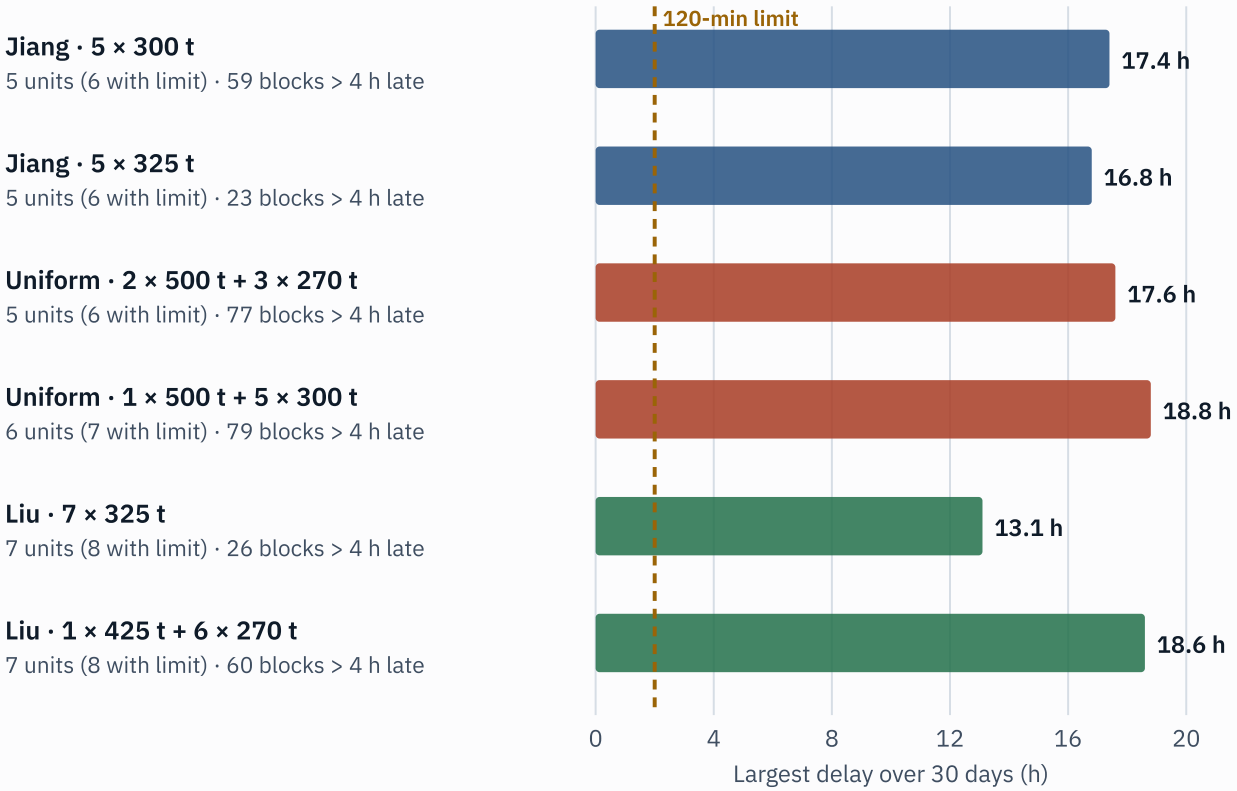
In 13 further production conditions, light units with a matched number of heavy units win 95.3% of 2,340 comparisons.

Heavy means 350 to 540 tons. From left to right: with no heavy blocks, a fleet of 300-ton vehicles is cheapest. With some, small vehicles plus one or two 550-ton units. With more, only 550-ton units. At every share, the cheapest fleet couples 3.3% of blocks or fewer on average.

Source: 30 further working days generated by this study, with heavy shares from 0 to 50%. The ranges of 350–540 t for heavy blocks and 100–300 t for the rest are assumptions. The chart uses the straight-line price fit and full-shift crews; all 9 vehicle-to-wage ratios give the same result.

Finding 4: service needs both an on-time rate and a delay limit

Without a delay limit, these fleets qualify with one vehicle fewer. The chart shows their latest block over 30 days.



6

fleets save one vehicle without a limit, at the cost of a latest block 13–19 h late.

The delayed blocks are mostly coupled ones. With only an average on-time rate, hard-to-carry blocks can be pushed back indefinitely.

- A 2-h limit raises the least cost by only 0–1.8%; a 30-min limit by at most 4.7%
- Without a limit, only 75% of winners couple at most one in ten under the Jiang and uniform masses; with any limit, 100% do

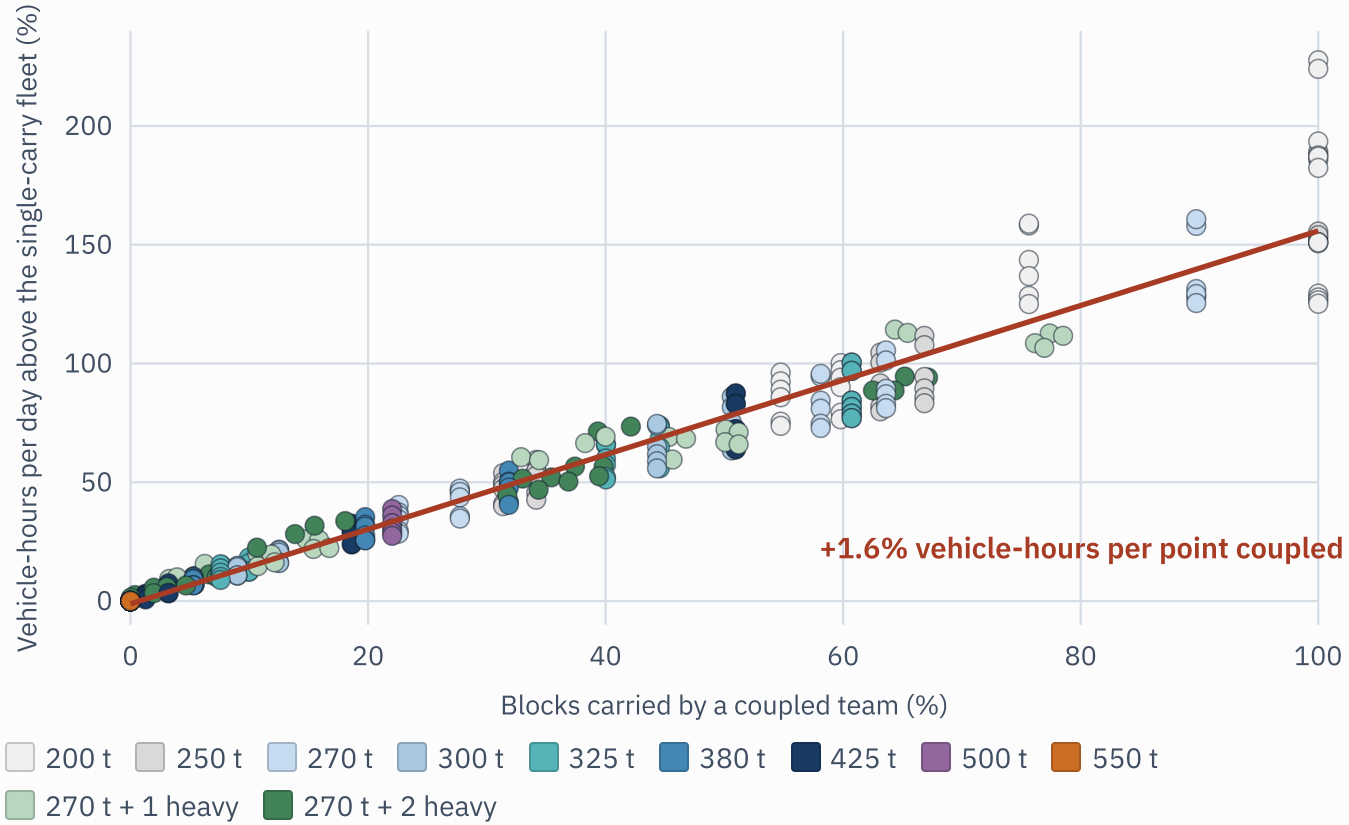
A delay limit costs almost nothing and rules out fleets that qualify by delaying.

With only a 95% target, a fleet can buy one vehicle fewer, but it pushes hard-to-schedule coupled blocks back by more than ten hours. It is like a courier that is 95% on time but leaves the rest for a day. A 2-hour limit adds at most 1.8% to cost.

Source: this study, three mass profiles with short handling, delay limits from 30 min to none.

Why coupling costs: several vehicles and crews are held for the whole trip

330 fleets (without the extra-heavy stress test): daily vehicle-hours above the single-carry fleet of the same condition



+1.6%

Each extra percentage point of blocks coupled adds 1.6% to the fleet’s daily vehicle-hours

2.2×

Fleets that couple more than 60% of blocks work 2.2 times the vehicle-hours of the single-carry fleet

90%

Of the extra time, a median of about 90% is the repeated trips and joining; waiting for partners is about 2%

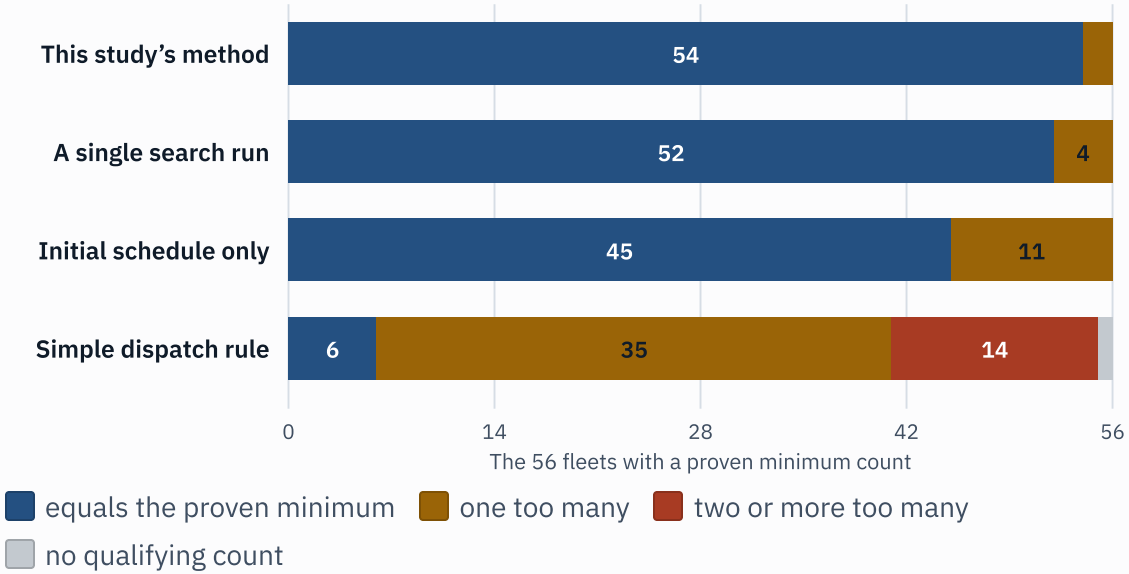
> **Every vehicle carries a crew, so a vehicle held is a crew held. With full-shift crews, light units can be cheaper only if coupling stays below about 2–18% of the workload.**

Each dot is one fleet. More coupling means more vehicle time, and about 90% of the extra is several vehicles driving the same trip. Every vehicle has a crew, so a busy vehicle means a busy crew. Extra vehicle time is extra labor.

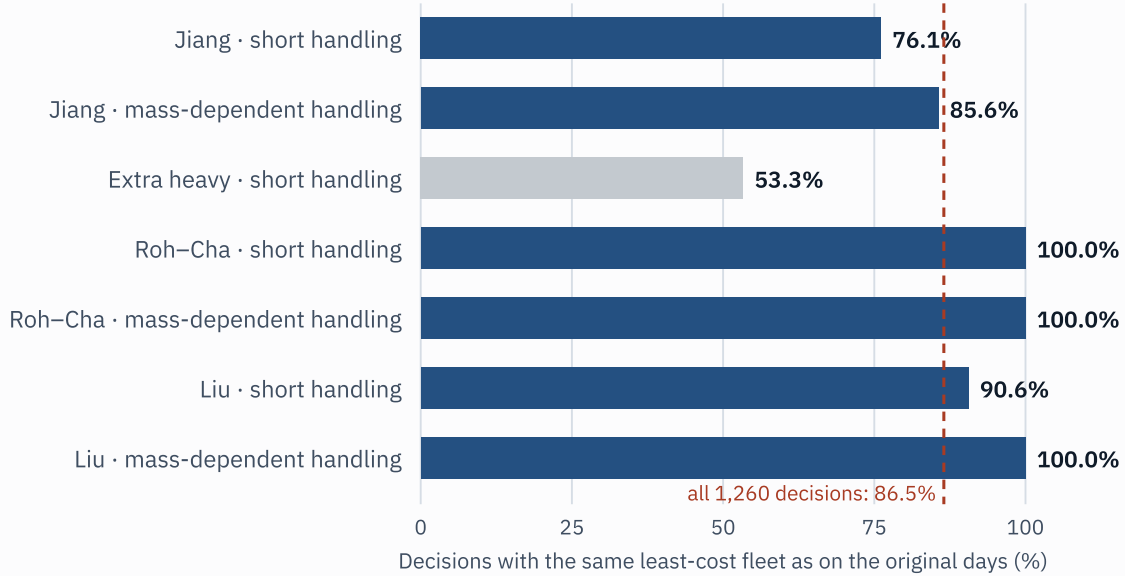
Source: this study, working and waiting time of every fleet, and a block-by-block breakdown.

Check 1: against exact solutions, and on new working days

Small instances (one batch of 12–13 blocks): counts found against the proven minimum (56 fleets)



Counts re-determined on 30 unused working days: share with the same cheapest fleet as before



54 / 56

fleets match the proven minimum and the other 2 are one over; all 360 comparisons pick the right cheapest fleet

39 / 42

fleets keep their count and the rest change by one; outside the stress test, 98.3% of cheapest fleets still couple at most one in ten

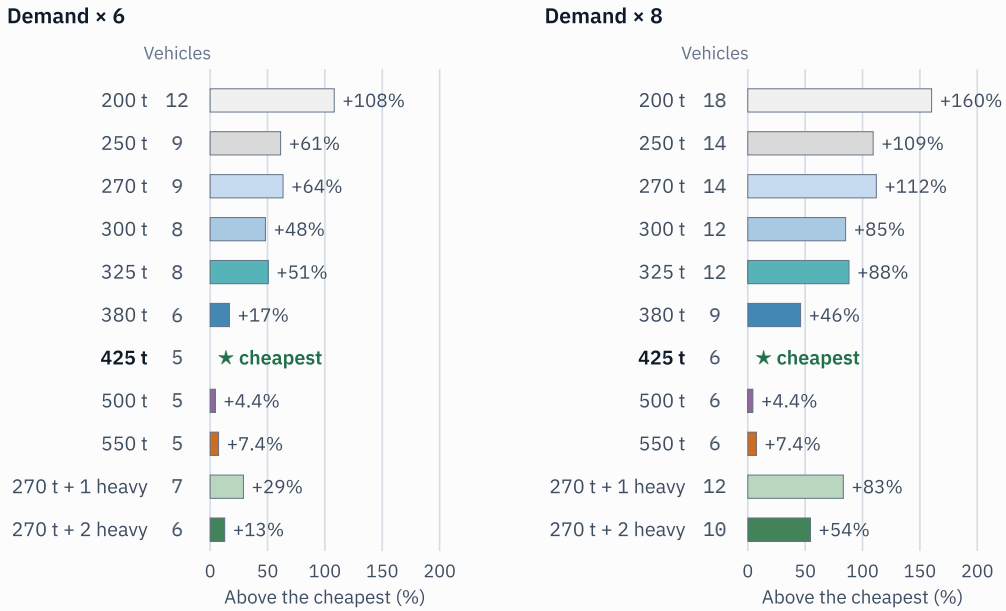
First, against exact answers: on small examples the fewest vehicles can be proven, and our method matches 54 of 56 fleets. Second, on 30 working days never used before, 39 of 42 fleets keep the same count.

Source: this study. Exact solutions from Google OR-Tools CP-SAT on 80 single-batch instances; exact solutions are only possible for single batches, so full-day counts are the smallest found. The fresh days cover 42 fleets in 7 production conditions.

Check 2: another shipyard and other parameters leave the finding unchanged

Second yard: public data from a Shanghai cruise-ship yard, demand scaled up 6 and 8 times. Bars show the cost above the cheapest fleet.

Vehicle-to-wage ratio 12.4



- **Second yard**: 5–6 units of 425 t suffice with no coupling, cheapest in 179 of 180 comparisons
- **Manufacturer speeds**: with each model's published speed, about 97% of cheapest fleets still couple at most one in ten
- **Unmeasured parameters**: changed one at a time, the choice moves only between neighboring sizes. With zero joining time a coupling fleet wins in one condition; from 2.5 min, low-coupling fleets win again
- **Handling time varying ±50%**: 145 of 360 comparisons change their cheapest fleet, but the new winners couple 0.1% of blocks or fewer
- **On-time target of 90% or 98%**: 40 and 37 of 42 fleets keep their count

At a cruise-ship yard in Shanghai, five or six 425-ton vehicles were enough, with no coupling, and that fleet was cheapest in 179 of 180 comparisons. With other speeds, joining times, handling times and on-time targets, the best size moved, but the cheapest fleet still hardly coupled.

Sources: second-yard data from Liu et al. (2022), Tables 4–5 and data S1 (PLOS ONE, CC BY). Demand multiples and the 10-h shift are assumptions. The rest is this study.

When does coupling help? The scope of this study

Four cases where coupling pays or is needed

- **No vehicle can carry the block alone**: coupling is the only way
- **Joining takes almost no time**: coupling fleets start to win only at zero joining time
- **One crew can run a whole coupled team**: coupling then saves the most labor
- **Vehicles cost far more than people**: a vehicle's daily purchase cost above about 1.4 crew-days, as with future remote-controlled or driverless vehicles

> **Under the conditions studied, the cheapest purchase favors single-vehicle capacity and keeps coupling for exceptional blocks.**

Scope

- Costs cover purchase and labor only, not maintenance, energy or resale value
- Routes are static shortest paths, without congestion or conflicts between vehicles
- Every coupled team is assumed physically feasible, which favors coupling. Road width, clearance and wheel loads also limit the heaviest single units; the net effect is not yet tested
- Demand is steady; seasonal peaks, renting and staged purchases are outside the scope
- Process routes and road paths are given inputs, not optimized
- Full-day counts are the smallest found, not proven minima

Coupling belongs where it fits: blocks no vehicle can carry alone, or cases where joining is almost free, one crew runs a whole team, or vehicles cost far more than people. Future remote or driverless vehicles may fall into the last group. The right column gives the scope of the findings.

Source: this study.

Five pieces of advice for planning a transporter fleet

1

Let one vehicle carry every common block

The cheapest is usually the smallest size that carries (nearly) every block alone, or light units with a few heavy ones. Larger vehicles save no further units.

2

Keep coupling within about 2–18% of the workload

Beyond it, the extra vehicles and crews outweigh the lower price of light units. Near it, daily timetables must decide.

3

Serve a few heavy blocks with a few heavy units

Set the number of heavy units by the heavy blocks' share of the workload. In 96.6% of the comparisons tested, this is within one unit of the cheapest choice.

4

For backup, compare a spare vehicle with coupling

Estimates with one vehicle's capacity removed make a second heavy unit or one more vehicle the cheapest backup. Real breakdown cover also needs a check of where the remaining vehicles are.

5

Set a delay limit and report near-cheapest options

With only an on-time rate, fleets delay hard blocks by hours. Options often differ by only 1–2% in cost, so list them together.

In one sentence: coupling is a remedy for exceptional blocks, not a purchasing strategy.

The five points come down to this: let one vehicle carry the common blocks, keep coupling for the few extra-heavy ones, add a delay limit to the on-time rule, and compare several options when costs are close.

Source: this study.

Conclusions: answers to the three questions

(1) What to buy

The smallest size that carries nearly every block alone; with few heavy blocks, light units plus one or two heavy ones. Larger vehicles save nothing more.

(2) When coupling pays

Only when no vehicle can carry the block, joining is near-instant, one crew runs a whole team, or vehicles cost far more than people.

(3) How to define service

Set both an on-time rate and a per-block delay limit. With only an on-time rate, fleets delay coupled blocks by more than ten hours.

Future work

- Check that coupled teams are physically feasible under road geometry and load distribution
- Optimize process routes together with the fleet
- Compare staged fleet expansion under seasonal demand

Status

The paper is complete and being prepared for submission. All 214,210 scheduling runs can be reproduced from the code.

Thank you. Questions and comments are welcome.

> The three answers in one line: buy vehicles that can carry blocks alone, keep coupling for special blocks, and set both an on-time rate and a delay limit.

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